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How do Low-Concentration Reactive Impurities in the CO_2 Stream Impact Geochemical Reactions in Chalk Reservoirs?

R. Mokhtari, K. Ghahramani, and K. L. Feilberg, Danish Offshore Technology Centre, Technical University of Denmark, 2800 Kgs. Lyngby, Denmark

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Abstract

Depleted oil and gas reservoirs in the Danish sector of the North Sea are composed primarily of chalk, a lithology distinguished by exceptionally high porosity, low permeability, and a mineralogical dominance of calcite. While these characteristics offer excellent storage capacity for CO_2, they also create susceptibility to chemical alteration when exposed to acidic fluids. This vulnerability is amplified when the injected CO_2 contains reactive impurities such as SO_2, NO_2, and H_2S, even at low concentrations typical of post-combustion CO_2 capture streams. These impurities, upon dissolution in formation brines, generate strong acids capable of accelerating calcite dissolution, altering pore structure, and potentially compromising mechanical stability. This study presents a detailed investigation into these effects using dynamic core flooding experiments conducted under reservoir-representative pressures and temperatures. Alternating cycles of impure CO_2 injection and brine flooding were designed to simulate transient conditions at advancing plume fronts. Effluent chemistry was monitored by ion chromatography, revealing that SO_2 induces rapid, high-magnitude dissolution, H_2S produces moderate but sustained effects, and NO_2 causes minimal direct dissolution but can trigger secondary precipitation under oxidizing conditions. Dissolution was confined primarily to brine–CO_2 contact zones, with negligible reaction in dry CO_2 regions. These findings emphasize that even trace impurities must be carefully controlled to ensure long-term storage security.

Keywords: CO_2 Storage, Chalk, Impurities, Geochemistry

Introduction

The storage of carbon in the form of compressed CO_2 within subsurface geological formations has emerged as one of the most promising and potentially cost-effective strategies for reducing the emission of heat-trapping CO_2 into the atmosphere. Over the past decades, carbon capture and storage (CCS) has been increasingly recognized as an indispensable and large-scale technology for achieving deep and sustained reductions in atmospheric CO_2 concentrations, particularly in the context of meeting ambitious climate targets (IPCC, 2023). CCS involves capturing CO_2 from major point sources, transporting it, and securely storing it in geological formations to prevent its release into the atmosphere. Among the range of geological storage options available, depleted oil and gas reservoirs occupy a special position. They are considered particularly suitable due to their proven sealing capacities, well-documented and well-understood structural configurations, and the presence of existing infrastructure that can be repurposed for CO_2 injection and monitoring (Mokhtari et al., 2014 and 2016).

In the Danish North Sea sector, many of these depleted hydrocarbon reservoirs are composed predominantly of Maastrichtian chalk formations. These chalk deposits are the product of extensive geological accumulation of coccolithophorid skeletons over millions of years. The resulting lithology is fine-grained, with a microporous calcite matrix that provides an unusually high storage capacity. However, this same structure, while advantageous for volume, also provides a large reactive surface area that makes the chalk particularly vulnerable to acid-induced alteration (Bergfors et al., 2020). The interplay between this high porosity, low permeability, and calcite-dominated mineralogy means that while chalk formations can host significant volumes of CO_2, they can also undergo chemical and mechanical changes when exposed to acidic fluids. The viability of any CO_2 storage scheme in such geological settings depends on a broad range of interrelated factors. These include, but are not limited to: CO_2-induced corrosion and scale formation in the reservoir and associated infrastructure; the availability and cost of CO_2 from industrial sources; the accessibility and condition of existing pipeline networks; the temperature and pressure conditions both at the surface and within the target formation; the sealing integrity of the overlying caprock; potential biological activity within the subsurface environment; the injectivity of the reservoir; the extent to which CO_2 can be trapped by mineralization processes; and the details of fluid–rock interactions over short and long timescales. Importantly, each of these aspects can be influenced by the presence of impurities in the injected CO_2 stream.

The chemical composition of the CO_2 stream depends directly on the characteristics of the fuel source and the specific capture technology employed. In practice, CO_2 with a certain level of impurities is far more commonly available in the volumes required for offshore transport and injection. This is because both capture processes and transportation systems inevitably lead to the inclusion of some contaminant species. Such impurities can originate from incomplete separation during capture, from the materials used in transport, or from reactions occurring along the pipeline route. Typical impurities present in industrial CO_2 streams include nitrogen (N_2), oxygen (O_2), argon (Ar), carbon monoxide (CO), sulfur oxides (SO_x), hydrogen sulfide (H_2S), nitrogen oxides (NO_x), and water (H_2O), as well as a range of other compounds such as hydrocarbons, amines, and degradation products of amines formed during amine-based capture processes (see Table 1). The relative concentrations of these impurities vary widely depending on the upstream processes.

Table 1—Impurities in CO_2 streams, with the typically most abundant highlighted in blue.

CO_2	H_2S	Hg	Terpenes
H_2O	CH_4	Ethanol	Amines
N_2	HCN	Methanol	Nitrosamines
O_2	NH_3	Glycol	Nitramines
SO_x	CO	Acetaldehyde	Particles
Ar	NO_x	Formaldehyde	Particulates (ash)
H_2	Cd	Aromatis	Oil
Cl	HCl	C_xH_y	

The fact that chalk reservoirs consist primarily of compressed carbonate sediment raises specific concerns about their storage safety and the long-term mechanical and chemical stability of the reservoirs under CO_2 injection. A major issue associated with injecting large amounts of CO_2 into such formations is the potential for structural damage caused by the acidic nature of CO_2. The primary mineral in chalk, calcite, is susceptible to dissolution in the presence of acids. When supercritical CO_2 is injected into a formation saturated with brine, it dissolves into the aqueous phase to form carbonic acid. This weak acid is capable of dissolving calcite, thereby altering pore geometry and potentially modifying both porosity and permeability (Hao et al., 2020). In actual CCS operations, CO_2 streams from capture facilities often contain low concentrations of reactive impurities such as sulfur dioxide (SO_2), nitrogen dioxide (NO_2), and hydrogen sulfide (H_2S). Once these impurities dissolve in formation brines, they yield stronger acids, sulphuric acid, nitric acid, and hydrosulphuric acid, which can significantly accelerate calcite dissolution compared to carbonic acid alone (Waldmann et al., 2016; de Visser et al., 2008). The specific impact of each impurity depends on its dissociation strength, solubility, and reactivity with chalk. For instance, SO_2 produces highly soluble and strongly acidic solutions, NO_2 can promote oxidation reactions that indirectly affect mineral stability, and H_2S, although producing weaker acids, can sustain dissolution effects over longer periods. These differences mean that the overall impact on reservoir integrity will vary depending on the composition of the injected CO_2 stream.

It is essential, therefore, to identify and evaluate the risks of undesired mechanical and chemical damage to the reservoir arising from fluid–rock interactions under realistic storage conditions. If carbonate reservoirs can maintain their mechanical stiffness and strength during CO_2 injection, or if any risks can be managed effectively, significant cost savings may be realized in the transportation and storage phases of CCS projects. The interaction between CO_2 and formation water leads to the formation of a weak acid that dissolves calcite minerals until a chemical equilibrium is reached. This raises fundamental questions about how CO_2 alters the chalk structure when mixed with saline formation waters. The stiffness and strength of chalk are controlled by the properties of the constituent grains and their bonding, which is often expressed as the cementation degree or Biot's effective stress coefficient (Hao et al., 2020). Studies have shown that SO_2 impurities can lead to the dissolution of minor dolomitic phases (Bolourinejad & Herber, 2014). The extent to which chemical dissolution results in mechanical weakening depends not only on the amount of calcite dissolved under a given injection scenario but also on whether the area of calcite grain contacts decreases, which would tend to weaken the rock, or increases, which could result in strengthening (Kim et al., 2018). Secondary mineral precipitation has been observed in some studies, with possible phases including nitrates and iron-bearing minerals (Wang et al., 2011). In many experimental observations, dissolution is concentrated in the localized zones where brine and CO_2 are in contact, with little to no reaction occurring in regions exposed only to dry CO_2 (Alam et al., 2014). Changes in grain contact area directly influence the mechanical behavior of chalk. Reductions in contact area generally lead to structural softening and weakening, whereas increases can produce strengthening—although this latter case may come at the cost of blocking smaller pore throats and reducing permeability (Raza et al., 2019). Experimental work involving the injection of supercritical CO_2 into chalk has, in some cases, revealed the formation of high-dissolution channels, known as wormholes, particularly near the injection point. Despite this, the majority of experiments have not documented significant pore blockage or measurable mechanical weakening over the timescales tested (Alam et al., 2014). However, it is important to note that experimental datasets remain limited, particularly with respect to the long-term mechanical consequences of calcite dissolution under realistic reservoir conditions. Factors such as the presence of micro-quartz, clays, reservoir temperature, and other mineralogical or operational variables can significantly influence outcomes.

From a geomechanical standpoint, the process of injecting CO_2 can present challenges both in the reservoir and in the associated wells. These challenges can negatively affect injectivity and may, in some cases, compromise well integrity (Madland et al., 2006; Torsæter & Cerasi, 2018). Excessive effective stresses, if not carefully managed, can result in borehole wall failure, particularly in sections of the well that are not mechanically protected, potentially leading to collapse. The solubility of CO_2 in formation water is a fundamental parameter in determining the storage capacity of a given host rock. Solubility is strongly influenced by both pressure and temperature, with higher pressures and lower temperatures generally increasing the amount of CO_2 that can dissolve. Salinity also plays a role, typically reducing CO_2 solubility, although the magnitude of this reduction varies depending on the composition of the dissolved salts, or background electrolyte (Liu et al., 2011). When CO_2 dissolves in brine, it lowers the pH, which can result in the dissolution of calcite and other minerals, as observed in experimental work (Pimienta et al., 2017). Under certain operational conditions, CO_2 injection can also lead to salt precipitation. For example, when dry CO_2 is injected, it can displace or vaporize water in the near-wellbore region. As the water evaporates, dissolved salts precipitate, occupying portions of the pore space. If these precipitated salts block CO_2 flow paths, the result can be an increase in injection pressures and a consequent reduction in injectivity (Yusof et al., 2022).

Impurities in the CO_2 stream further complicate the picture. They can raise both the bubble point and dew point pressures, necessitating higher operational pressures to maintain CO_2 in a single-phase state. If these pressures are not maintained, the system may transition into a two-phase regime, which can lead to unstable flow patterns such as slug flow, abrupt pressure surges, and operational challenges for compressors and pumps. Moreover, the presence of a liquid phase in combination with water and reactive impurities such as SO_x, NO_x, and H_2S leads to acid formation, including carbonic, sulfuric, and nitric acids, which greatly increases corrosion rates in pipelines and equipment. Impurities also have significant effects on the thermophysical properties of CO_2 mixtures. Non-condensable gases, for example, tend to reduce density and increase the critical pressure of the mixture, both of which can reduce storage efficiency and influence injectivity in geological formations. Nitrogen, even at relatively low concentrations, has one of the most pronounced effects on phase transitions, while methane exerts a more subdued influence. Water is especially problematic, not only because it is immiscible with dense-phase CO_2, but also because, above its solubility limit, it forms a free aqueous phase. In the presence of acidic gases, this aqueous phase becomes highly corrosive, accelerating the degradation of infrastructure. Water also contributes to the formation of gas hydrates, which can block flow paths within pipelines and possibly within the reservoir itself. Other impurities such as SO_x, NO_x, and H_2S, even in trace amounts, can react with water to form corrosive acids or produce elemental sulfur deposition. Such deposition can clog pore networks in the storage formation. While some more condensable impurities, such as SO_2, can increase the density of the CO_2 mixture and thus potentially enhance storage capacity, their corrosive potential remains a significant concern for both infrastructure and reservoir performance.

Experimental Section

The geochemical effects, including precipitation and dissolution of identified impurities, are tested on real core samples. Core flood injection experiments are carried out in high pressure experimental setups that allow for changing the temperature to investigate the effects of temperature on any chalk dissolution caused by lower or higher $CaCO_3$ solubility caused by impurities. The core flood experiments are carried out at the temperature and pressure of the target depleted chalk reservoirs using the relevant impurities.

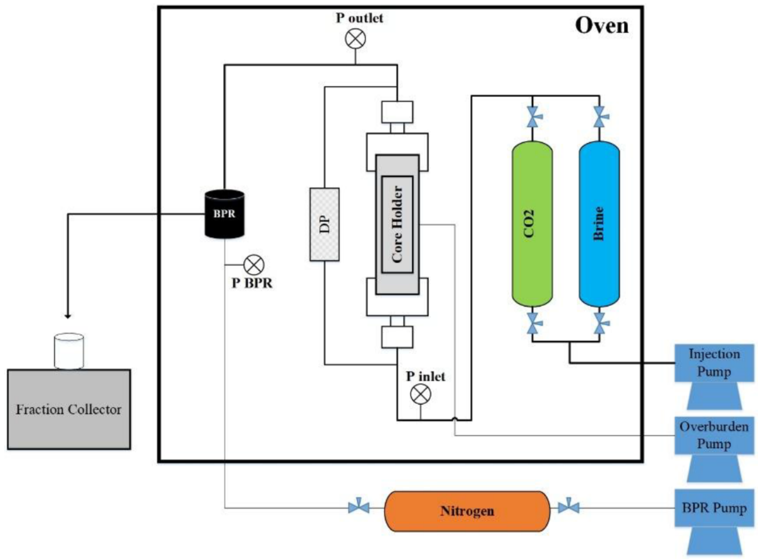

Figure 1—Detailed schematic of the core flooding apparatus, showing the configuration of high-precision liquid and gas delivery pumps, the axial core holder with confining pressure, temperature control systems, and backpressure regulation. This arrangement enabled precise replication of subsurface pressure–temperature conditions.

Chalk core plugs (>95% calcite) were obtained from a depleted oil field in the Danish North Sea and machined to dimensions suitable for core flooding experiments. Petrophysical characterization confirmed porosities between 30–35% and permeabilities of 0.3–0.4 mD, representative of reservoir chalk. Synthetic brines matching formation water, seawater, and diluted seawater compositions were prepared using analytical-grade reagents to replicate in situ ionic strengths and compositions.

Impure CO_2 mixtures were generated by blending high-purity CO_2 with precise concentrations of SO_2 (200–600 ppm), NO_2 (200–400 ppm), or H_2S (300–500 ppm) using mass flow controllers. The core flooding system comprised high-accuracy pumps, an axial core holder equipped with a confining pressure jacket, thermocouples for temperature monitoring, and a backpressure regulator to maintain 100 bar. The system was thermostatically controlled to 35 °C, ensuring fluid properties matched reservoir conditions.

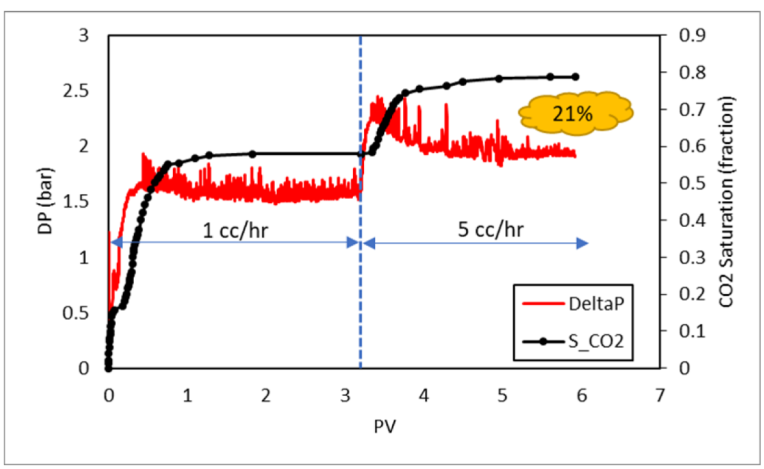

Figure 2—CO_2 injection into a formation water-saturated chalk core plug.

Experiments simulated advancing CO_2–brine interfaces through alternating cycles of brine injection and impure CO_2 flooding. Flow rates of 0.1 cc/min represented reservoir-like conditions. Effluent brine samples were collected in airtight vials to prevent degassing and analyzed by ion chromatography for Ca^{2+}, Mg^{2+},

Na^+, Cl^-, and SO_4^{2-}. Sodium served as a conservative tracer for normalizing data, compensating for dilution or concentration effects (Liu et al., 2011).

Results

For SO_2-containing CO_2 injections, Ca^{2+} concentrations in the effluent increased sharply within the first pore volumes, reaching peaks more than twice those of pure CO_2 runs. Mg^{2+} release was also observed, indicating dissolution rates remained high for several pore volumes before plateauing, suggesting either surface passivation or depletion of reactive sites.

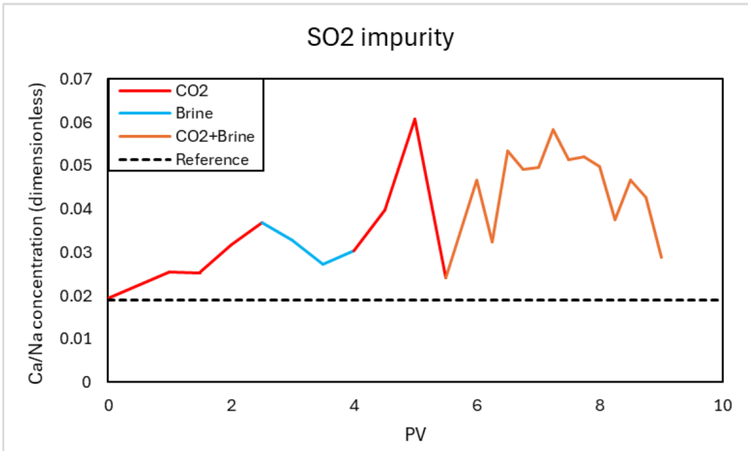

Figure 3—Calcium release profiles during core flooding with a SO_2-containing CO2 injection stream. The data are normalized with respect to sodium to show the Calcium relative to total salinity in the effluent. These datasets highlight both the intensity and kinetics of dissolution, with SO_2 generating steep early rises in Ca^{2+} concentration.

H_2S-containing runs exhibited slower initial Ca^{2+} increases (Fig. 4) but sustained dissolution over extended injection periods, indicating a more gradual but persistent reactivity. This could be due to the conversion kinetics of H_2S to acidic species. The total Ca^{2+} release was observed to be lower than for SO_2, reflecting a lower tendency to dissolve dolomitic components. However, the fact that Mg^{2+} was still seen to be released over the relatively short time scale of the experiment indicates that it needs to be considered for longer-term storage scenarios. NO_2-containing runs displayed only minor changes in Ca^{2+}(Fig. 5).

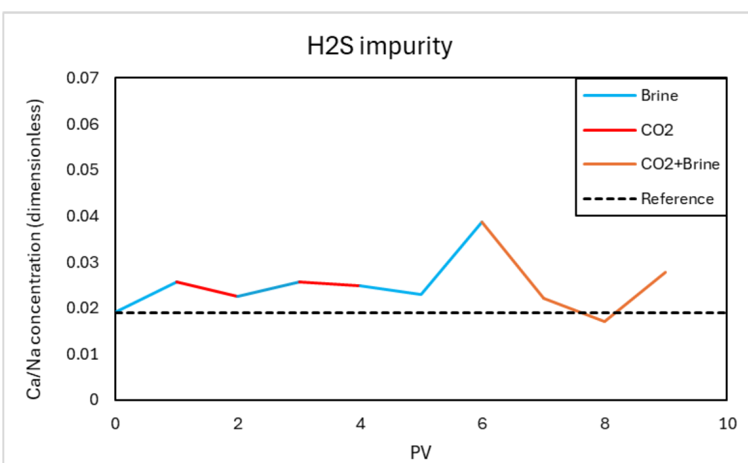

Figure 4—Calcium release profiles during core flooding with an H_2S-containing CO_2 injection stream. The data are normalized with respect to sodium to show the Calcium relative to total salinity in the effluent.

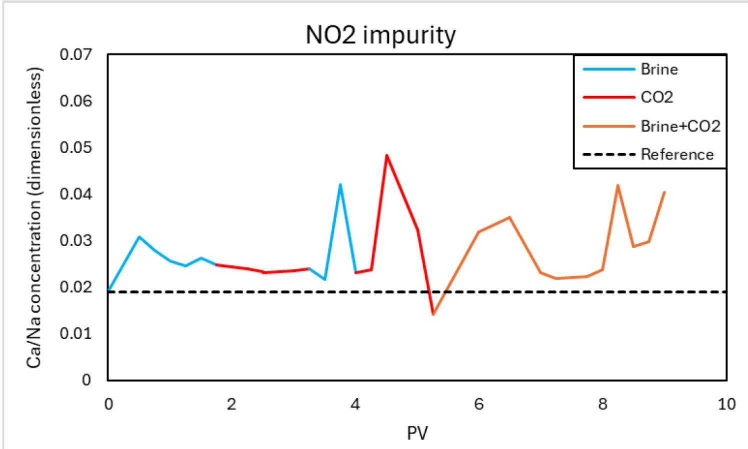

Figure 5—Calcium release profiles during core flooding with a NO_2-containing CO2 injection stream. The data are normalized with respect to sodium to show the Calcium relative to total salinity in the effluent.

Discussion

The observed dissolution behavior reflects the combined influence of impurity solubility, acid strength, and reaction kinetics. SO_2's high solubility and strong acidity upon hydration drive rapid calcite dissolution, particularly in the early stages of exposure. H_2S produces weaker acids, resulting in slower dissolution but prolonged reactivity. NO_2's limited solubility and oxidative nature likely promote precipitation reactions that can offset dissolution effects.

The plateauing dissolution trends point to possible surface passivation by secondary mineral coatings or reductions in reactive surface area. Such phenomena may stabilize dissolution rates but can also create heterogeneous alteration patterns, potentially affecting local permeability. Even at low concentrations, these impurities have the potential to significantly modify near-wellbore geochemistry, altering porosity and permeability in ways that could influence injectivity and containment over time. Mitigation strategies include rigorous purification of CO_2 streams to remove reactive impurities, operational adjustments to limit brine saturation in critical zones, and chemical buffering of formation waters. Predictive models that couple geochemical and geomechanical processes (Raza et al., 2019) are essential for assessing the long-term implications of impurity-driven dissolution and for designing monitoring protocols.

Conclusions

This study confirms that chalk formations are highly sensitive to reactive impurities in injected CO_2, even at concentrations in the hundreds of ppm. SO_2 is the most aggressive, producing rapid and extensive calcite dissolution that can undermine rock stability. H_2S, while less aggressive, maintains dissolution over longer timescales, potentially leading to cumulative changes possibly due to conversion steps to sulphate. NO_2 shows a smaller but still measurable dissolution effect over the shorter timescale of the experiment. In addition, the impurities can cause precipitation effects, which can cause reservoir changes. Such localized effects can have significant operational impacts, especially in the near-wellbore environment. The fact that the dissolution reaches a plateau is promising for a longer-term storage scenario, however future studies should include the presence of bulk crude oil as well as flow experiments. Maintaining strict control over CO_2 purity, combined with careful reservoir management and continuous monitoring, is essential for ensuring the safe and effective use of chalk formations in CCS.

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